

SUSHANT KULKARNI

AI Engineer & Backend Developer

sushant.kulkarni841@gmail.com • +917775062359 • Pune,India

www.linkedin.com/in/sushantkulkarni841
http://sushantkulkarni.me/

PROFESSIONAL SUMMARY

AI Engineer & Backend Developer with 3+ years of experience building production-grade AI applications and scalable backend systems using Node.js, NestJS, Python, and AWS. Experienced in designing LLM-powered solutions, RAG pipelines, AI agents, and multimodal workflows while developing high-performance REST APIs, microservices, and cloud-native architectures. Strong expertise in system design, backend optimization, database design, and deploying secure, scalable applications for enterprise environments.

SKILLS

Languages:	Python, Typescript, Javascript
Backend:	Node.js, Nest.js, Express.js, Fastapi, Git, Docker, Microservices, Webhooks, WebSocket, OpenSearch, GraphQL, Kafka, Prisma ORM, Redis
Frontend:	React
AI & ML:	Large Language Models (GPT-5, Azure OpenAI, Claude, Llama) LLM, Retrieval-Augmented Generation (RAG), Prompt Engineering, Neo4j graph database, Vector Databases(ChromaDB, Weaviate, Pinecone), Generative AI, Model Context Protocol (MCP), Natural Language Processing (NLP), AI Agents, LangChain, LangGraph, Multimodal AI, Semantic Search, Prompt Chaining, Function Calling, TensorFlow, Scikit-learn, LLM Evaluation Frameworks, N8n
Databases:	PostgreSQL, DynamoDB
Cloud/DevOps:	AWS (EC2, S3, RDS, Lambda, API Gateway, IAM, CloudWatch), Azure (OpenAI, App Services, SQL, Blob Storage), Jira

WORK EXPERIENCE

Software Engineer | Team Lead — Centralogic Consultancy Pvt Ltd

Feb 2024 – Present

Pune, India

- Architected and deployed 3+ production-grade intelligent applications leveraging NestJS, Python, and AWS serverless primitives, scaling scalable microservices to process 10,000+ complex healthcare conversations.
- Led 0-to-1 system architecture and project execution for 2 core initiatives, directing PostgreSQL database design, REST API orchestration, and cloud infrastructure from concept to production launch.
- Implemented LoRA fine-tuning on open-source LLMs to specialize models for medical data parsing, substantially improving domain accuracy from 89% to 96%and clinical entity extraction.
- Engineered multi-stage RAG pipelines and autonomous agentic workflows featuring dynamic tool-use and function calling powered by LangChain, LlamaIndex, Pinecone, and ChromaDB.
- Developed a low-latency AI interview agent utilizing WebSockets, Node.js event-driven architecture, and real-time streaming APIs across Anthropic Claude, OpenAI GPT-4o, and Mistral models.
- Integrated computer vision pipelines for live candidate proctoring, leveraging OpenCV and InsightFace for real-time face detection, facial verification and multi-face tracking with more than 90% accuracy.
- Enforced structured payload validation using Pydantic and established dynamic, cost-aware model routing to optimize token usage and minimize API overhead expenses.
- Integrated LangSmith for end-to-end LLM observability, distributed tracing, and prompt evaluation, systematically tracking performance bottlenecks and eliminating output drift.
- Automated end-to-end healthcare intake workflows, reducing manual processing effort by ~80% through structured parsing, Redis caching, and asynchronous queue management via SQS.
- Built a streamlined candidate onboarding application in Next.js, optimizing frontend-backend integration to cut verification turnaround time from ~1 hour to ~20 minutes.
- Containerized scalable backend services using Docker, AWS Lambda, API Gateway, and S3, establishing automated CI/CD pipelines and maintaining 99.9% system availability.

Nashik, India

- Co-founded and built a development initiative delivering web and backend solutions for multiple clients.
- Designed and developed end-to-end applications, including system architecture, backend services, and deployment.
- Built scalable backend systems and REST APIs using Node.js for real-world business applications.
- Collaborated with clients to translate business requirements into effective technical solutions.

PROJECTS

Agent Onboarding System

- Architected a production-grade full-stack platform using NestJS and React.js, automating onboarding for 5,000+ agents.
- Reduced end-to-end processing time by 66% (60 → 20 minutes) by engineering automated validation pipelines and streamlined approval workflows.
- Engineered scalable backend services using Prisma ORM and PostgreSQL, Kafka optimizing schemas and queries for high-performance data retrieval.
- Secured the platform with Okta, JWT-based authentication, and Role-Based Access Control (RBAC) for multi-level authorization.
- Improved system performance and reliability using Redis for distributed caching and session management, maintaining sub-second API response times under concurrent load.
- Designed modular backend services using Object-Oriented Design and wrote Jest unit tests to achieve 90%+ test coverage, ensuring reliable code quality.

Technologies: Nest.js, React, PostgreSQL, AWS

Interview Agent

- Architected an AI-powered interview agent using Mistral, LLaMA, and GPT-4, enabling dynamic, context-aware conversations for technical role assessments.
- Engineered a multi-model orchestration pipeline to optimize response latency and cost-efficiency while maintaining high output quality across 1,200+ live interviews.
- Developed a high-performance backend using Python and FastAPI to manage conversation state, prompt chaining, and real-time streaming responses.
- Integrated a multimodal stack with Whisper (large-v3) for Speech-to-Text (STT), Deepgram for low-latency Text-to-Speech (TTS), and InsightFace for real-time biometric proctoring.
- Built vision-based AI workflows for automated resume evaluation and candidate scoring, triggering interview pipelines based on profile analysis.
- Orchestrated event-driven workflows for scheduling, evaluation scoring, and automated candidate feedback notifications.

Technologies: Python, Fastapi, RAG, LLM, Multimodal Processing, Node.js, CosmosDB, Azure

Scribing AI

- Developed an AI-powered medical scribe platform, building LLM pipelines and backend services using Python, FastAPI, Node.js, Express.js, and PostgreSQL to transcribe doctor-patient conversations and generate structured clinical summaries.
- Built end-to-end pipelines to extract medical findings and generate SOAP (Subjective, Objective, Assessment, Plan) notes for physicians.
- Processed 10,000+ healthcare conversations with ~96% accuracy in extracting clinically relevant insights.
- Fine-tuned Whisper models with 400K+ trainable parameters using LoRA.
- Engineered scalable APIs, transcript-processing workflows, database integrations, and real-time report generation pipelines.
- Reduced manual documentation effort by automating clinical note generation and standardizing reporting workflows.

Technologies: Python, FastAPI, LLMs, Speech-to-Text (STT), Prompt Engineering, REST APIs, PostgreSQL, Fine-tuning

EDUCATION

MET BKC College of Engineering , Nashik, India

Jul 2020 – Jun 2024

Bachelors of Engineering in Information Technology (GPA: 8.88/10)

CERTIFICATIONS

AWS Certified Cloud Practitioner (ID: b49df5fb1eaa46bcb6d8ca7f6b2e859e)
Amazon Web Services

Apr 2026 – Apr 2029